NETWORKWALKS

Cybersecurity Lab Environment Setup

Kali Linux virtual lab — technical project report

Prepared by Madhurja Bhattacharjee
Batch ID B084N
Project Week 1 · Project Module 1
Host Windows 11
Hypervisor Oracle VirtualBox 7.2.8
Guest Kali Linux 2026.3

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Executive Summary

This report documents the setup and verification of a Kali Linux cybersecurity lab running as a virtual machine on a Windows 11 host. The work establishes a repeatable base environment, configures a dedicated VirtualBox NAT Network, assigns a static guest IPv4 address, checks routing, external connectivity and DNS resolution, validates shared-folder access, and records a baseline snapshot.

The documented results support the conclusion that the Kali guest had working connectivity to its configured gateway and an external IP address, successfully resolved a domain name, and read a test file from the configured shared folder. Clipboard and drag-and-drop were enabled in VirtualBox settings, but a separate functional transfer test was not recorded.

Environment at a Glance

Item	Recorded configuration
Host operating system	Windows 11
Hypervisor	Oracle VirtualBox 7.2.8r173730
Guest operating system	Kali Linux 2026.3
VM resources	2 GB RAM · 2 virtual CPUs · 128 MB video memory
Network mode	NAT Network — NetworkWalks-NAT
Network range	10.0.0.0/24 · DHCP disabled
Guest IPv4 / gateway	10.0.0.2/24 · 10.0.0.1
DNS resolver	8.8.8.8
Shared folder	Shared-Files → /downloads
Baseline snapshot	NetworkWalks-Kali-Base-Setup

Implementation Status

Status	Items
Configured and verified	Static IPv4, routing table, gateway ping, external IP ping, DNS lookup, shared-folder file read, baseline snapshot.
Configured; functional test not recorded	Bidirectional shared clipboard and drag-and-drop.
Planned / not completed in this phase	Android-x86 VM, Windows 10 VM, inter-VM connectivity testing, Wireshark browser-based labs.

Important Configuration Notes

- The VirtualBox VM display name contains an older image label (2026.1), while the installed guest operating system reports Kali Linux 2026.3. The report distinguishes the VM label from the guest OS version.
- The host shared-folder directory is intentionally redacted in the configuration record; the share name and guest mount point are documented.
- A NAT Network provides address translation and connectivity; it should not be treated as a complete security-isolation boundary. Host firewall policy and guest configuration remain relevant.
- The interface state string UNKNOWN, as displayed by ip -br addr, is not by itself proof of failed connectivity. The recorded gateway and external connectivity tests succeeded.
- The verification text file is a curated record of previously observed results, not a fresh command execution log.

01 — Project Objectives and Scope

1. Purpose

The purpose of this project is to establish a functional virtual cybersecurity laboratory using Oracle VirtualBox and Kali Linux.

The environment provides a foundation for subsequent cybersecurity exercises involving networking, reconnaissance, traffic analysis, vulnerability assessment, and security testing.

2. Primary Objectives

– Install and verify Oracle VirtualBox.

– Import and configure the Kali Linux virtual machine.

– Create a dedicated VirtualBox NAT Network.

– Assign a static IPv4 address to Kali Linux.

– Configure the default gateway and DNS server.

– Verify local gateway connectivity.

– Verify external network connectivity.

– Verify DNS resolution.

– Configure shared-folder integration between Windows and Kali.

– Enable bidirectional clipboard and drag-and-drop settings.

– Create a baseline VM snapshot.

– Maintain screenshot evidence and technical documentation.

3. Implemented Scope

The current completed scope covers the Kali Linux base environment and its accompanying technical documentation.

Component	Status
VirtualBox installation	Completed
Kali Linux VM configuration	Completed
NAT Network	Completed
Static IP configuration	Completed
Internet connectivity	Verified
DNS resolution	Verified
Shared-folder integration	Verified
Clipboard configuration	Enabled
Drag-and-drop configuration	Enabled
Baseline snapshot	Created
Technical documentation	Completed
Consolidated technical report and evidence gallery	Completed

4. Planned Extensions

The following are planned activities and are not part of the completed Kali setup:

• Android-x86 VM deployment.

- Windows 10 VM deployment.
- Inter-VM communication tests.
- Wireshark practical exercises.

5. Expected Outcomes

At the end of the current phase, the lab should provide:

- A bootable Kali Linux VM.
- A predictable static IPv4 configuration.
- Working gateway and external connectivity.
- Functional DNS resolution.
- Verified host-to-guest shared-folder access.
- A recoverable baseline snapshot.
- Reproducible configuration documentation.

6. Authorization and Safety

All future security testing must remain within the designated lab environment or other explicitly authorized systems.

02 — Prerequisites

1. Host Requirements

The lab was configured on a Windows 11 host.

Component	Requirement / Actual Configuration
Host OS	Windows 11
Hypervisor	Oracle VirtualBox
VirtualBox version	7.2.8r173730
Host memory	8 GB
Virtualization	Hardware virtualization enabled
Storage	Sufficient free space for VM files and evidence

An 8 GB host has limited memory headroom. Avoid running multiple resource-intensive virtual machines simultaneously.

2. Software Requirements

- Oracle VirtualBox.
- Kali Linux VirtualBox image.
- Windows PowerShell.
- Git for version control.
- A web browser for downloading official software and accessing documentation.

3. Guest VM Requirements

Setting	Actual Value
Guest OS	Kali Linux 2026.3
VM memory	2048 MB
Virtual CPUs	2
Video memory	128 MB
3D acceleration	Disabled
Network adapter	PCnet-FAST III
Network attachment	NAT Network

The VM's displayed name retains an older 2026.1 image label. The installed guest operating system reports Kali Linux 2026.3.

4. Network Requirements

Parameter	Value
NAT Network	NetworkWalks-NAT
Subnet	10.0.0.0/24
Gateway	10.0.0.1
Kali static IP	10.0.0.2/24

Parameter	Value
DNS server	8.8.8.8
DHCP	Disabled

5. Shared Folder Requirements

Host directory:

`<host-shared-folder-directory>`

VirtualBox shared-folder name:

`Shared-Files`

Guest mount point:

`/downloads`

The shared folder is configured for automatic mounting and permanent availability.

6. Evidence Requirements

The project maintains screenshots for:

- VirtualBox version.
- NAT Network configuration.
- VM motherboard, processor, and display settings.
- Network adapter settings.
- Clipboard and drag-and-drop settings.
- Shared-folder configuration.
- Static IP configuration.
- Routing and device status.
- Connectivity tests.
- DNS resolution.
- Shared-folder access.
- Snapshot creation.

7. Safety Notes

Use official software sources wherever possible.

Do not expose credentials, private keys, personal files, or sensitive host information in public screenshots or repository commits.

03 — VirtualBox Setup

1. Hypervisor

Oracle VirtualBox was installed and verified on the Windows 11 host.

Installed version: 7.2.8r173730

2. Kali VM

The Kali Linux VirtualBox appliance was imported and configured.

Setting	Configuration
VM name	kali-linux-2026.1-virtualbox-amd64
Guest OS	Kali Linux 2026.3
Base memory	2048 MB
Processors	2
Video memory	128 MB
3D acceleration	Disabled

The VM name reflects the downloaded image's older naming convention. The installed guest reports version 2026.3.

3. Network Adapter

The first network adapter was configured as follows:

Parameter	Value
Adapter	Adapter 1
Attached to	NAT Network
Network name	NetworkWalks-NAT
Adapter type	PCnet-FAST III (Am79C973)
Promiscuous mode	Deny
Cable connected	Enabled

Other network adapters were disabled.

4. Shared Folder

The host directory was configured as a VirtualBox shared folder.

Setting	Value
Host path	<host-shared-folder-directory>
Share name	Shared-Files
Guest mount point	/downloads
Auto-mount	Enabled
Permanent	Enabled

The guest's `vboxsf` module was verified as loaded, and the Kali user was added to the `vboxsf` group.

A host-created test file was successfully read from the guest.

5. Integration Settings

The following VirtualBox integration options were enabled:

- Shared clipboard: Bidirectional.
- Drag and drop: Bidirectional.

These are configuration settings. A separate functional clipboard or drag-and-drop transfer test was not recorded.

6. Baseline Snapshot

Snapshot name:

`NetworkWalks-Kali-Base-Setup`

The snapshot preserves the configured Kali baseline before additional lab activities.

7. Evidence

Relevant screenshots are stored in:

`evidence/screenshots/kali/`

See the evidence inventory in `06-verification.md`.

04 — Kali Linux Configuration

1. Guest Operating System

The configured guest operating system is Kali Linux 2026.3.

The VirtualBox VM name retains the downloaded image's older 2026.1 naming convention.

2. VM Hardware

Component	Configuration
Base memory	2048 MB
Virtual CPUs	2
Video memory	128 MB
3D acceleration	Disabled
Network adapter	PCnet-FAST III
Network mode	NAT Network

3. Network Interface

The primary network interface is:

`eth0`

The static IPv4 address assigned to the interface is:

`10.0.0.2/24`

The connection profile used is:

`Wired connection 1`

4. Static Network Configuration

The following NetworkManager commands were used:

```
sudo nmcli connection modify "Wired connection 1" \
  ipv4.method manual \
  ipv4.addresses 10.0.0.2/24 \
  ipv4.gateway 10.0.0.1 \
  ipv4.dns 8.8.8.8
```

The connection was restarted:

```
sudo nmcli connection down "Wired connection 1"
sudo nmcli connection up "Wired connection 1"
```

5. Shared Folder Integration

The host shared folder is mounted at:

`/downloads`

The guest was checked for the VirtualBox shared-folder module:

```
lsmod | grep vboxsf
```

The user group membership was checked with:

```
groups
```

The `vboxsf` group was present.

A test file created on the Windows host was accessible from Kali:

```
ls -la /downloads
```

```
cat /downloads/shared-folder-test.txt
```

Expected test-file content:

```
NetworkWalks shared-folder test
```

The content was successfully read from the guest.

6. Clipboard and Drag-and-Drop

VirtualBox General > Advanced settings:

Feature	Setting
Shared clipboard	Bidirectional
Drag and drop	Bidirectional

These settings were enabled. No separate end-to-end transfer test is claimed.

7. Snapshot

The following baseline snapshot was created:

`NetworkWalks-Kali-Base-Setup`

This provides a recovery point before additional lab changes.

05 — Network Configuration

1. Network Design

The lab uses a VirtualBox NAT Network.

Parameter	Configuration
Network name	NetworkWalks-NAT
Network address	10.0.0.0/24
Gateway	10.0.0.1
Kali address	10.0.0.2/24
DHCP	Disabled
DNS	8.8.8.8

2. NAT Network Creation

The NAT Network was created from the Windows host using VBoxManage:

```
& $VBox natnetwork add `
    --netname "NetworkWalks-NAT" `
    --network "10.0.0.0/24" `
    --enable `
    --dhcp off
```

The network configuration was inspected with:

```
& $VBox natnetwork list
```

The network was configured with DHCP disabled.

3. Kali Static IPv4 Configuration

The Kali network connection was configured using NetworkManager:

```
sudo nmcli connection modify "Wired connection 1" \
  ipv4.method manual \
  ipv4.addresses 10.0.0.2/24 \
  ipv4.gateway 10.0.0.1 \
  ipv4.dns 8.8.8.8
```

The connection was restarted:

```
sudo nmcli connection down "Wired connection 1"
sudo nmcli connection up "Wired connection 1"
```

4. Interface Verification

Command:

```
ip -br addr
```

Recorded interface configuration:

```
eth0    UNKNOWN    10.0.0.2/24
```

5. Routing Verification

Command:

```
ip route
```

Recorded routes:

```
default via 10.0.0.1 dev eth0 proto static metric 100
10.0.0.0/24 dev eth0 proto kernel scope link src 10.0.0.2 metric 100
```

6. Gateway Connectivity

Command:

```
ping -c 4 10.0.0.1
```

Result:

- 4 packets transmitted.
- 4 packets received.
- 0% packet loss.
- Average round-trip time: approximately 0.752 ms.

7. External Connectivity

Command:

```
ping -c 4 8.8.8.8
```

Result:

- 4 packets transmitted.
- 4 packets received.
- 0% packet loss.
- Average round-trip time: approximately 45.137 ms.

8. DNS Resolution

Command:

```
nslookup google.com
```

The configured DNS server was `8.8.8.8`.

The lookup returned IPv4 and IPv6 address records.

9. Network Security Note

A VirtualBox NAT Network provides guest networking through NAT and allows connected VMs to communicate according to their network configuration.

It should not be described as a complete isolation or security boundary. Host firewall configuration and the behavior of connected guests remain relevant.

06 — Verification and Evidence

1. Purpose

This document records the checks performed after configuring the Kali Linux virtual machine and its network environment.

Results below reflect the observed lab configuration and command outputs.

2. Verification Summary

No.	Verification	Result
1	VirtualBox version	Verified
2	NAT Network	Verified
3	Kali motherboard settings	Verified
4	Kali processor settings	Verified
5	Kali display settings	Verified
6	Network adapter	Verified
7	Clipboard and drag-and-drop settings	Enabled
8	Shared-folder configuration	Verified
9	Static IP configuration	Verified
10	IP routing and interface status	Verified
11	Gateway and external connectivity	Passed
12	DNS resolution	Passed
13	Shared-folder file access	Passed
14	Baseline snapshot	Created

3. Network Test Results

3.1 Interface Address

Command:

```
ip -br addr
```

Observed IPv4 address:

`10.0.0.2/24`

Interface:

`eth0`

3.2 Routing Table

Command:

```
ip route
```

Observed default route:

```
default via 10.0.0.1 dev eth0
```

3.3 Gateway Ping

Command:

```
ping -c 4 10.0.0.1
```

Result:

- 4/4 packets received.
- 0% packet loss.

3.4 External IP Ping

Command:

```
ping -c 4 8.8.8.8
```

Result:

- 4/4 packets received.
- 0% packet loss.

3.5 DNS Resolution

Command:

```
nslookup google.com
```

Result:

- DNS server: 8.8.8.8.
- DNS lookup successful.
- IPv4 and IPv6 records returned.

4. Shared Folder Verification

The Windows host test file:

`shared-folder-test.txt`

was visible inside Kali under:

`/downloads`

The file content was successfully read:

```
NetworkWalks shared-folder test
```

The `vboxsf` module and group membership were also checked.

5. Integration Settings

The VirtualBox shared clipboard and drag-and-drop options were configured as Bidirectional.

A separate end-to-end transfer test was not recorded. Therefore, only the configuration is marked verified.

6. Snapshot Verification

Snapshot:

`NetworkWalks-Kali-Base-Setup`

The snapshot was created after the base configuration and verification steps.

7. Screenshot Evidence Inventory

Location:

`evidence/screenshots/kali/`

Evidence ID	Filename	Description
K01	K01-virtualbox-version.jpeg	VirtualBox version
K02	K02-nat-network.png	NAT Network settings

Evidence ID	Filename	Description
K03a	K03a-kali-motherboard.jpeg	VM motherboard settings
K03b	K03b-kali-processor.png	VM processor settings
K03c	K03c-kali-display.png	VM display settings
K04	K04-kali-network-adapter.png	Kali network adapter
K05	K05-clipboard-dragdrop.jpeg	Clipboard and drag-and-drop settings
K06	K06-shared-folder-configuration.png	Shared-folder configuration
K07	K07-static-ip-configuration.png	Static IP configuration
K08	K08-ip-route-device-status.png	Interface and routing status
K09-K10	K09-K10-connectivity-tests.png	Gateway and external connectivity
K11	K11-dns-resolution.png	DNS resolution
K12	K12-shared-folder-access.png	Shared-folder access
K13	K13-kali-base-snapshot.png	Baseline snapshot

The evidence set contains 14 image files because K03 is represented by three screenshots and K09/K10 share one combined screenshot.

8. Pending Verification

The following activities are outside the completed Kali base setup:

- Android-x86 deployment and connectivity.
- Windows 10 deployment and connectivity.
- Inter-VM communication.
- Wireshark practical exercises.

These will be documented after the corresponding tests are completed.

07 — Troubleshooting

1. Purpose

This document records useful diagnostic commands and troubleshooting guidance for the Kali Linux VirtualBox lab.

Only issues and solutions actually encountered should be described as completed fixes. The guidance below also includes general diagnostic procedures for future use.

2. Network Interface Not Showing Expected IP

Inspect interfaces:

```
ip -br addr
```

Inspect connection profiles:

```
nmcli connection show
```

Inspect active connections:

```
nmcli device status
```

Check the configured profile:

```
nmcli connection show "Wired connection 1"
```

If the connection is inactive, restart it:

```
sudo nmcli connection down "Wired connection 1"
sudo nmcli connection up "Wired connection 1"
```

3. Gateway Connectivity Failure

Check the routing table:

```
ip route
```

Check gateway reachability:

```
ping -c 4 10.0.0.1
```

Confirm:

- The VM network adapter is enabled.
- The adapter is attached to `NetworkWalks-NAT`.
- The NAT Network is enabled.
- The guest has the correct static IP.
- The default gateway is `10.0.0.1`.

4. Internet Connectivity Failure

First test the gateway:

```
ping -c 4 10.0.0.1
```

Then test an external IP:

```
ping -c 4 8.8.8.8
```

If the gateway works but the external IP does not, inspect the VirtualBox NAT Network configuration and host network connectivity.

5. DNS Resolution Failure

Inspect DNS configuration:

```
cat /etc/resolv.conf
```

Test DNS resolution:

```
nslookup google.com
```

The configured DNS server for this lab is:

`8.8.8.8`

6. Shared Folder Not Accessible

Check whether the VirtualBox shared-folder module is loaded:

```
lsmod | grep vboxsf
```

Check group membership:

```
groups
```

Check the mount directory:

```
ls -la /downloads
```

Confirm that the VirtualBox shared folder uses:

- Share name: `Shared-Files`
- Mount point: `/downloads`
- Auto-mount: enabled
- Permanent: enabled

If group membership has just changed, log out and log back in before testing again.

7. Clipboard or Drag-and-Drop Not Working

Inspect VirtualBox settings under:

Settings > General > Advanced

Check:

- Shared Clipboard: Bidirectional.
- Drag and Drop: Bidirectional.

These settings may depend on Guest Additions compatibility and the running guest session.

Enabling the settings alone does not prove that file transfer or clipboard synchronization works. Perform a separate functional test before claiming it is verified.

8. Low Host Memory

The host has 8 GB of RAM.

Running multiple virtual machines simultaneously can cause performance problems.

Recommended precautions:

- Run one resource-intensive guest at a time.
- Close unnecessary applications.
- Monitor host memory usage.
- Avoid increasing VM memory without checking available host resources.

9. General Diagnostic Commands

```
ip -br addr
ip route
nmcli device status
nmcli connection show
ping -c 4 10.0.0.1
ping -c 4 8.8.8.8
nslookup google.com
lsmod | grep vboxsf
groups
```

10. Documentation Practice

For every new issue:

– Record the observed symptom.

– Capture the relevant command output.

– Identify the likely cause.

– Apply one change at a time.

– Repeat the relevant verification test.

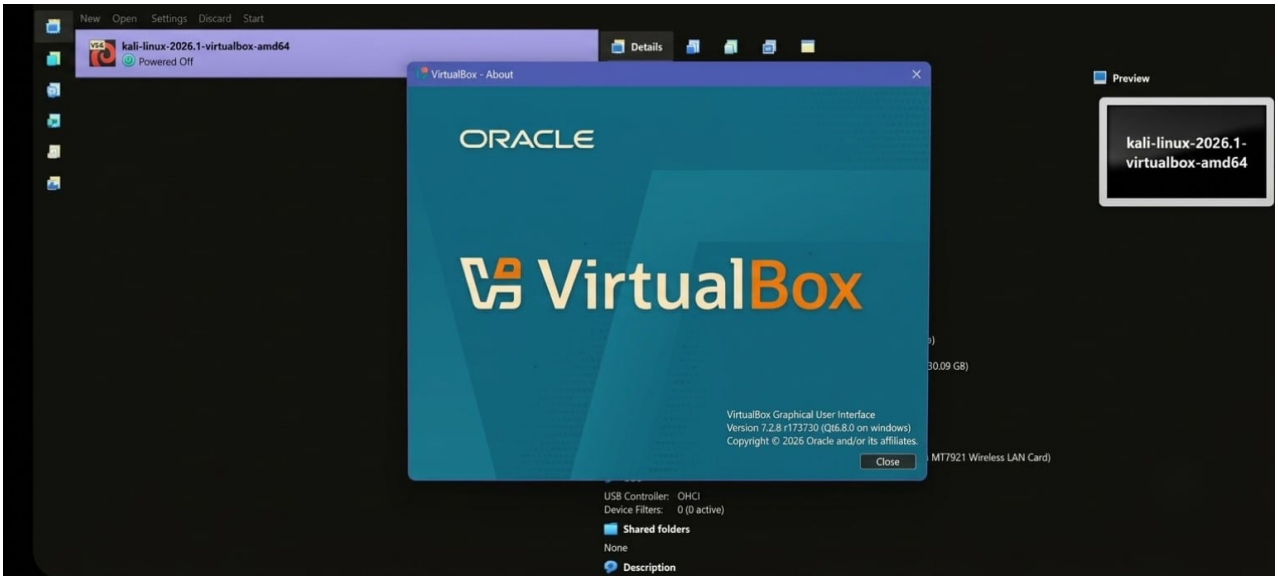
– Record the actual result.

Do not document a troubleshooting fix as successful unless the result has been verified.

Evidence Gallery

The following images are the project evidence supplied in evidence/screenshots/kali/. Captions describe what each image records; they do not imply additional tests beyond those documented.

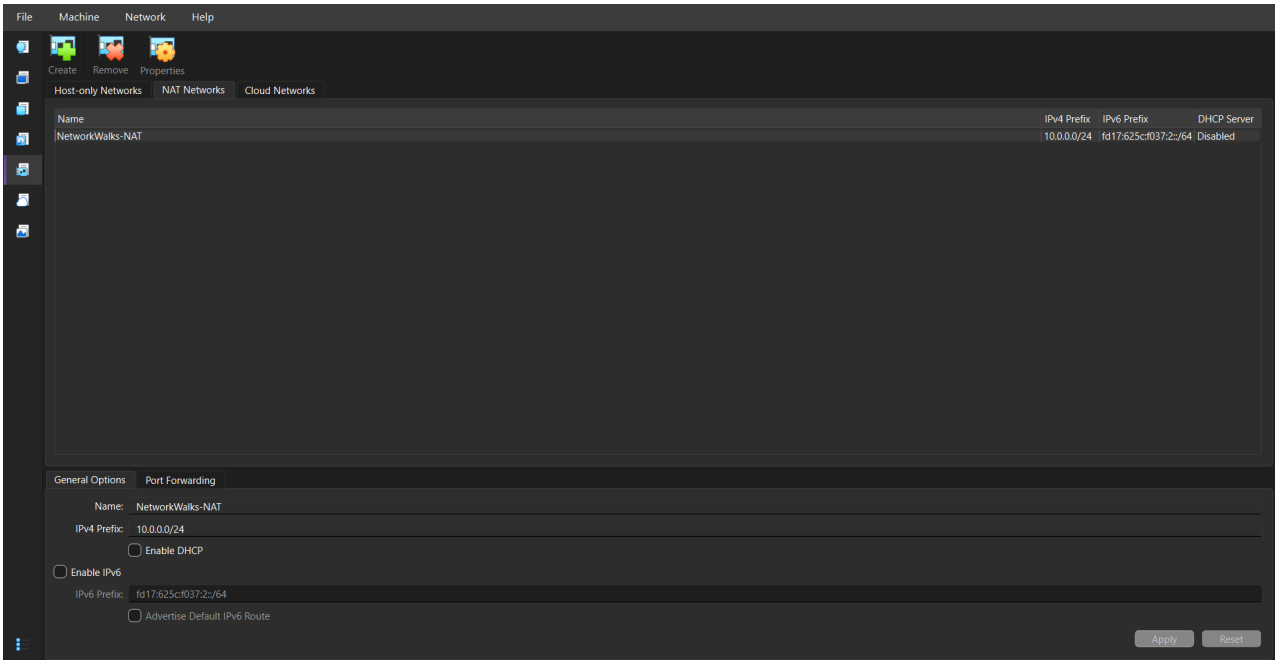
K01 — Oracle VirtualBox version



Host hypervisor version recorded for the lab environment.

`K01-virtualbox-version.jpeg` · 1536 × 691 px

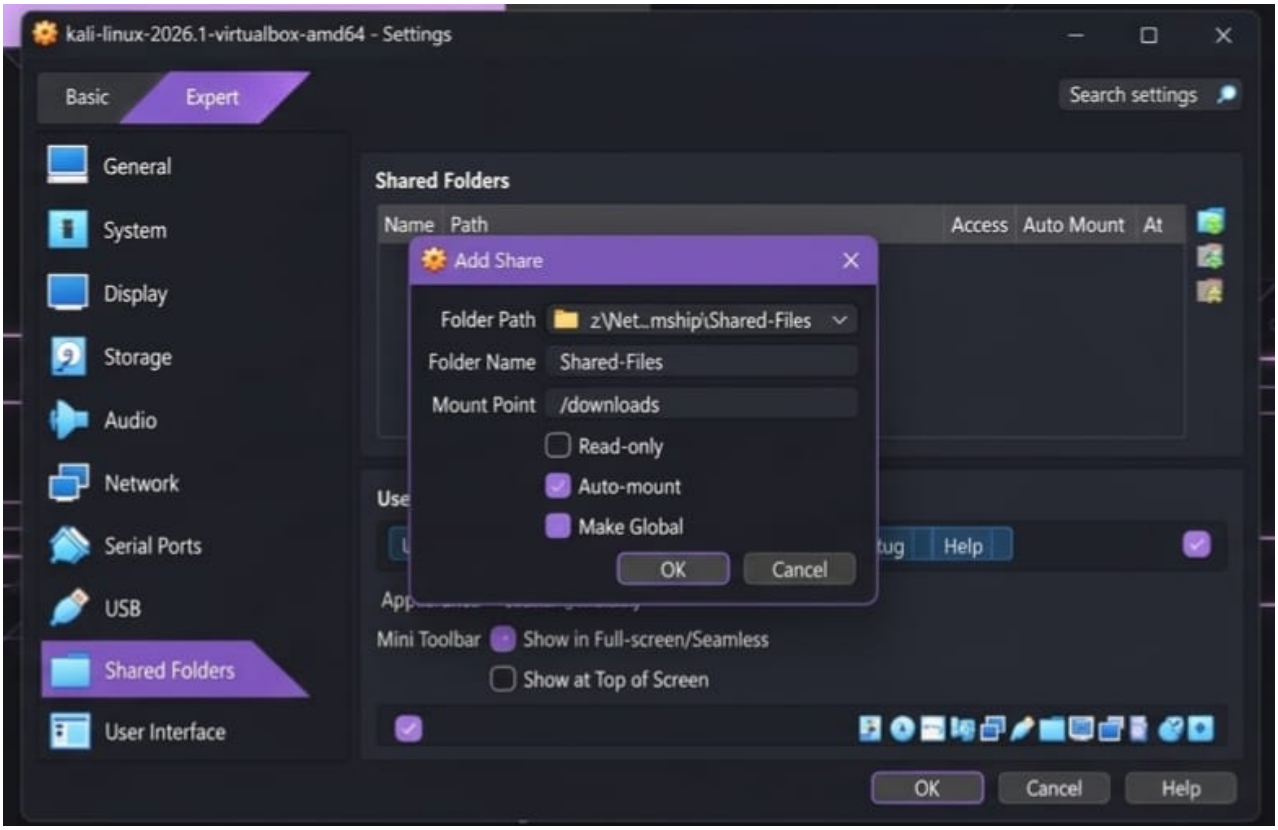
K02 — NAT Network configuration



NetworkWalks-NAT network configuration, including subnet and DHCP setting.

`K02-nat-network.png` · 1920 × 991 px

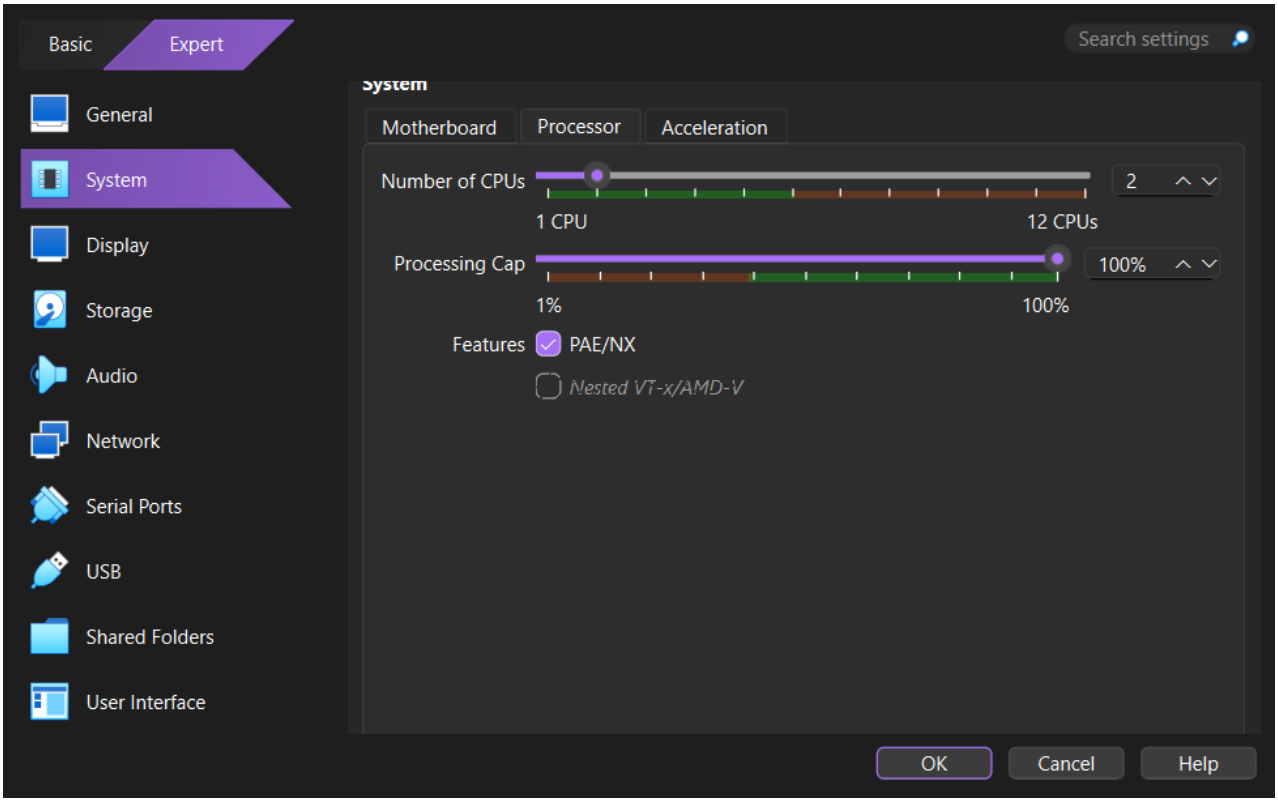
K03a — Virtual machine base hardware



Kali virtual machine motherboard and base configuration.

`K03a-kali-motherboard.jpeg` · 781 × 505 px

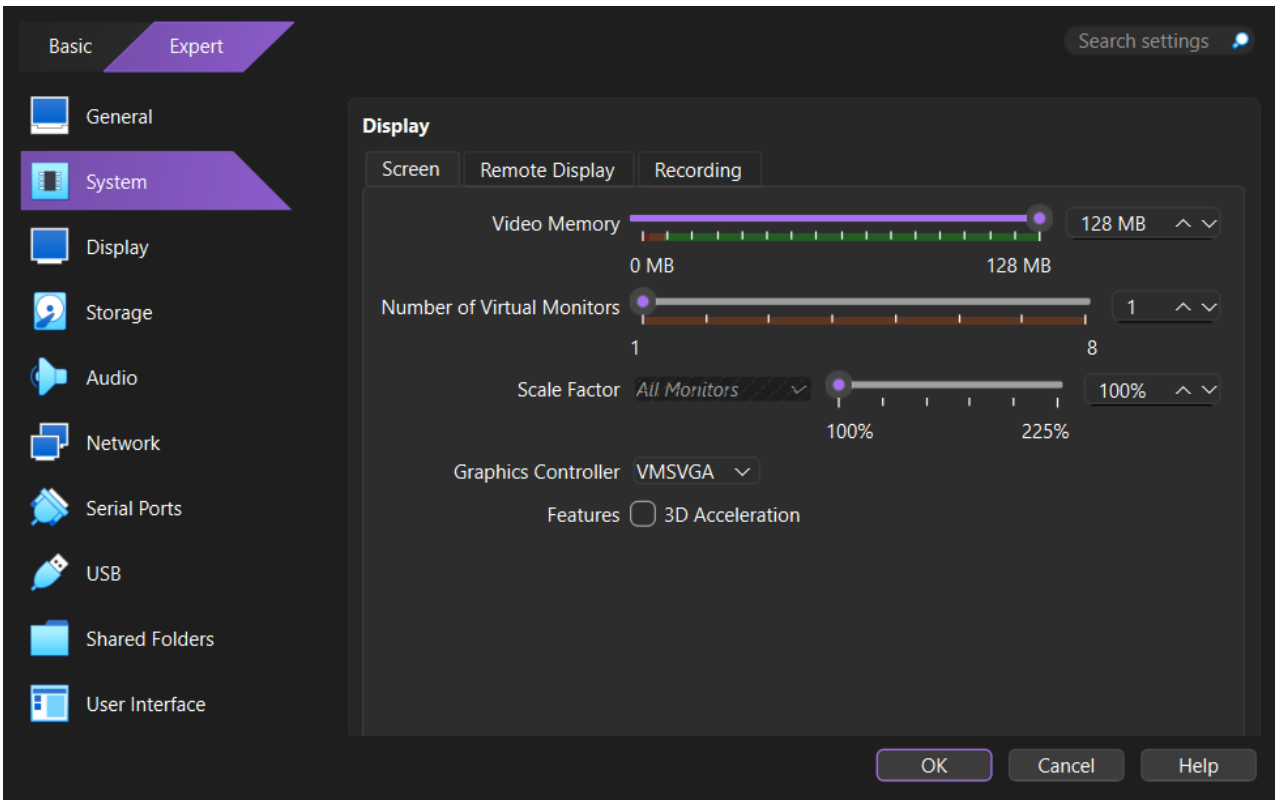
K03b — Virtual processor allocation



Two virtual CPUs assigned to the Kali guest.

`K03b-kali-processor.png · 974 × 608 px`

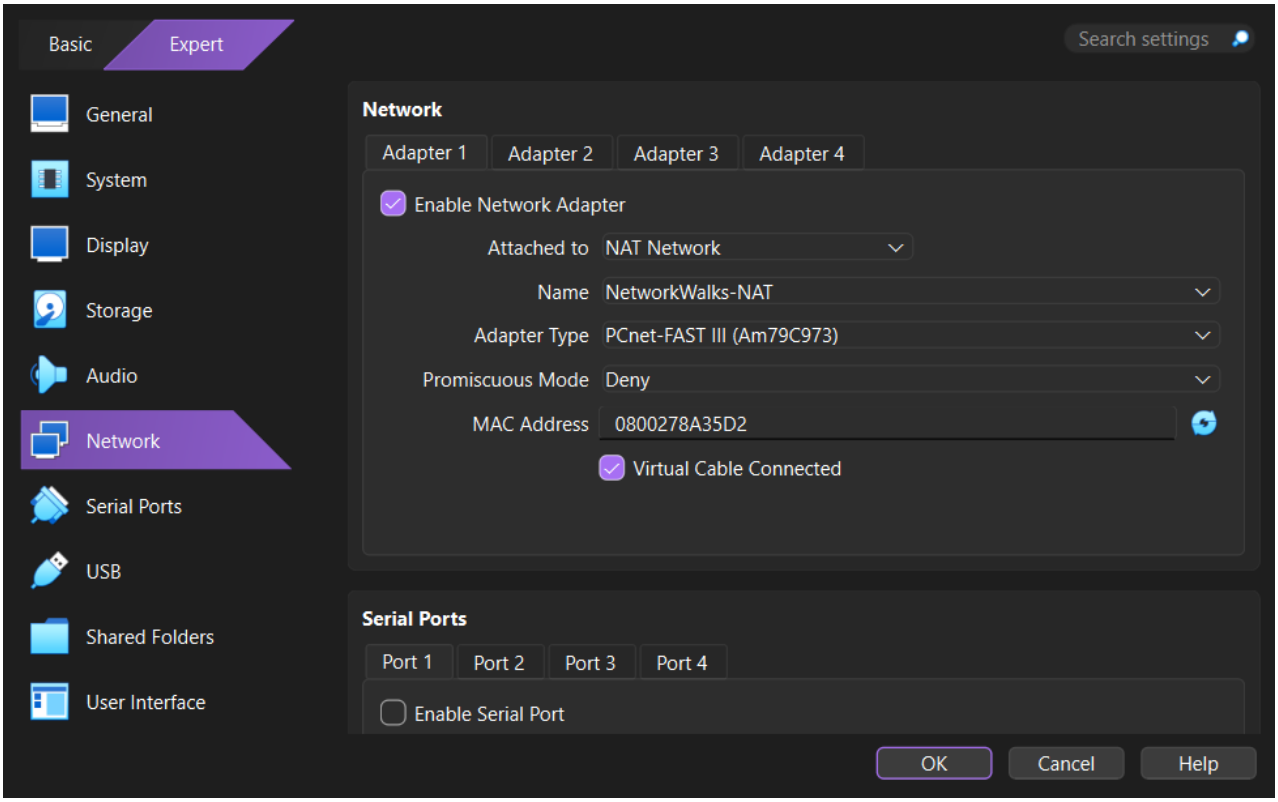
K03c — Virtual display configuration



Video memory and display acceleration configuration.

`K03c-kali-display.png` · 974 × 608 px

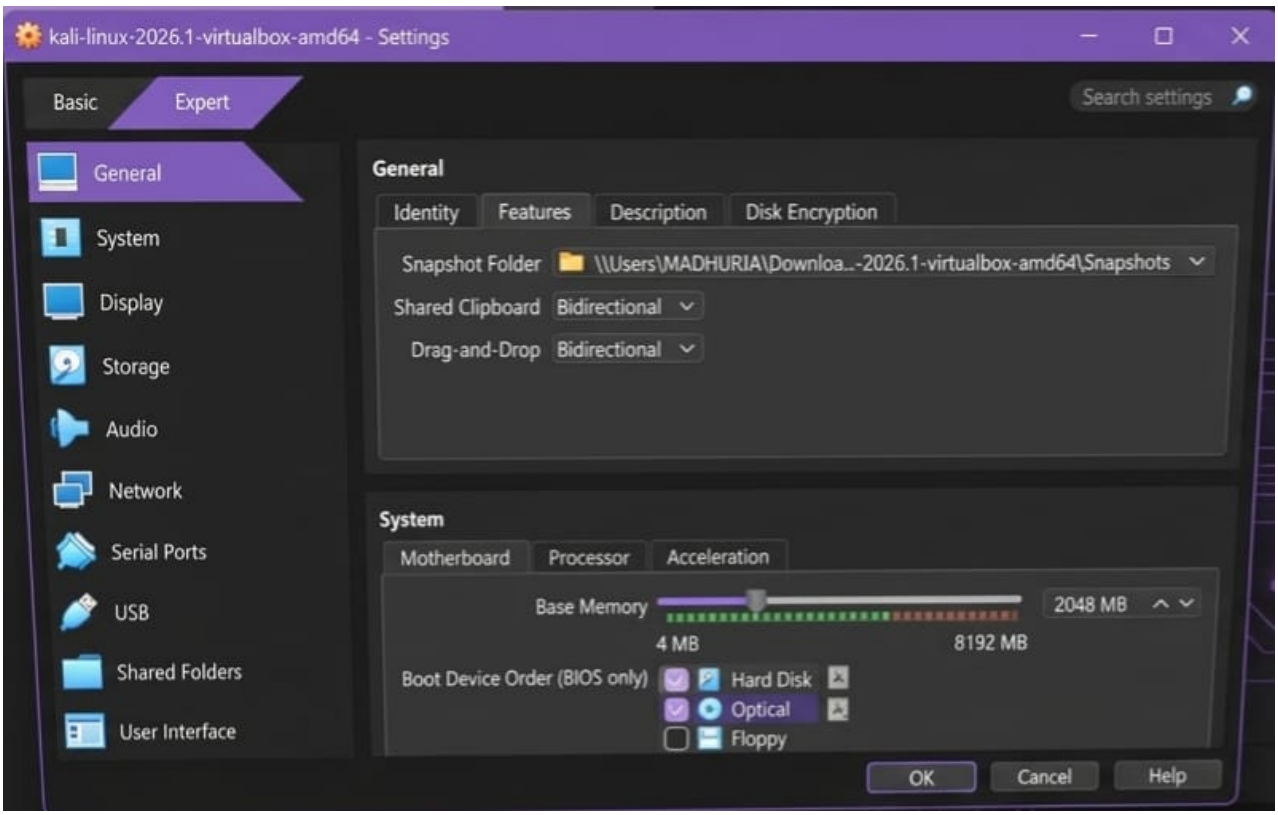
K04 — Virtual network adapter



Adapter model and NAT Network attachment used by the Kali guest.

`K04-kali-network-adapter.png · 974 × 608 px`

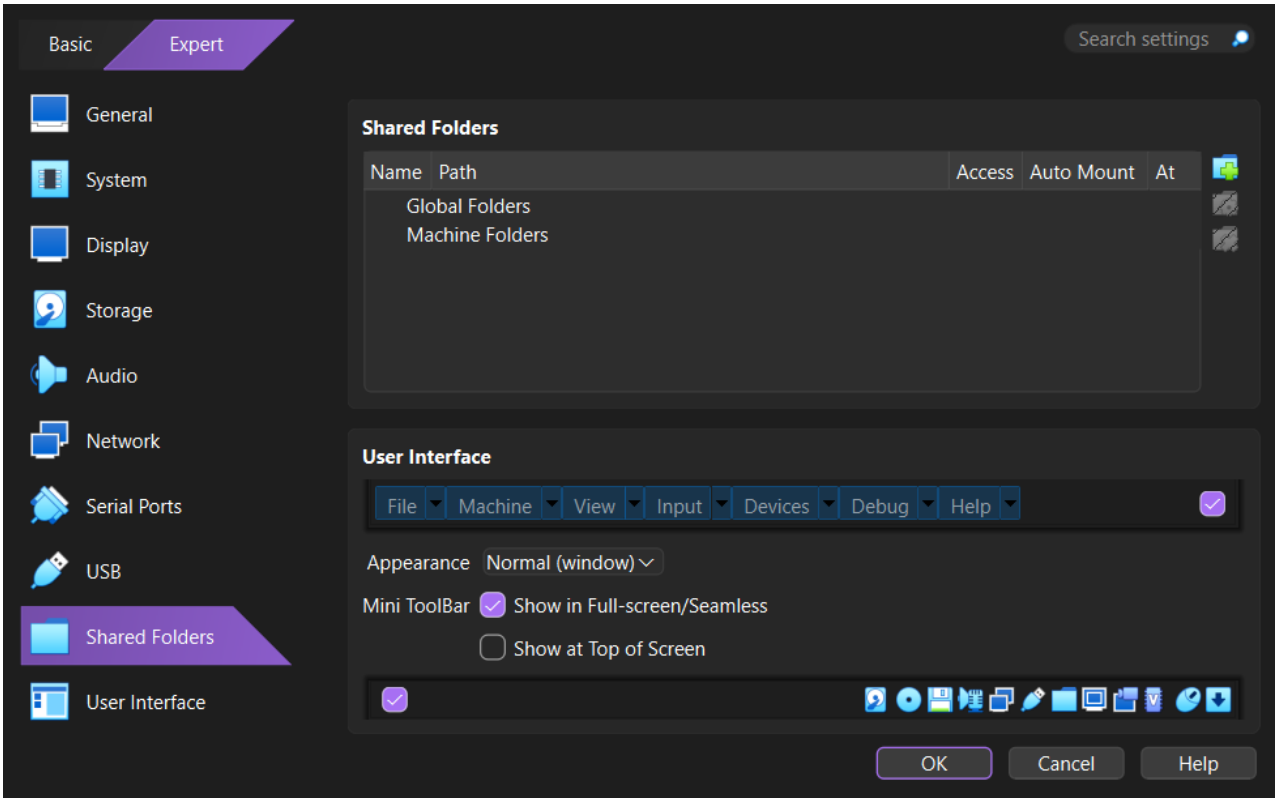
K05 — Integration settings



Bidirectional shared clipboard and drag-and-drop settings. These settings were configured; a separate functional transfer test was not recorded.

`K05-clipboard-dragdrop.jpeg` · 798 × 505 px

K06 — Shared folder configuration



Shared-Files host share and guest mount configuration.

`K06-shared-folder-configuration.png · 974 × 608 px`

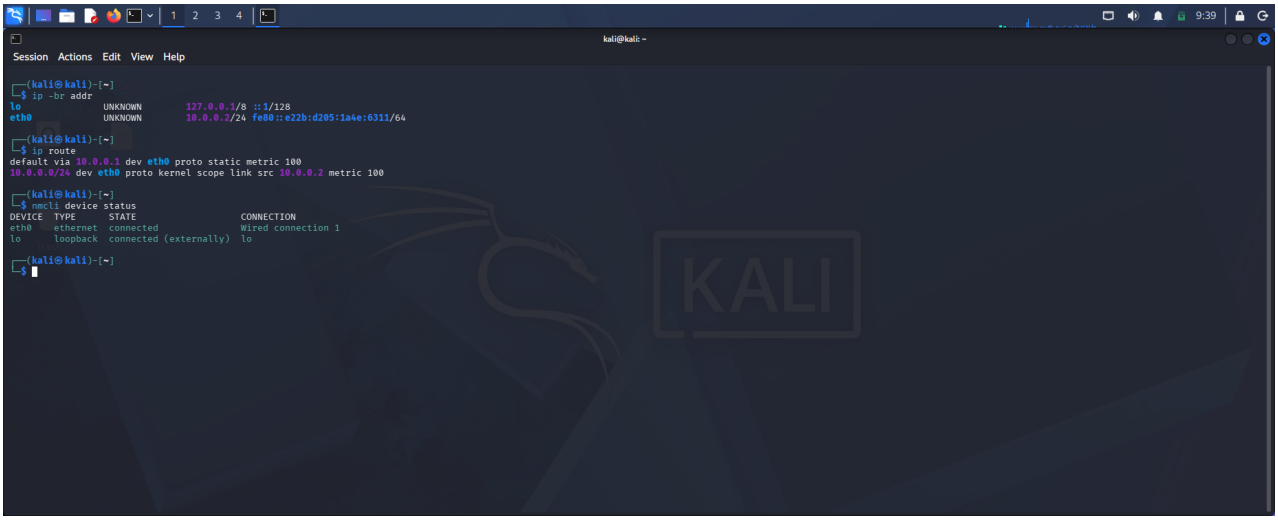
K07 — Static IPv4 configuration



NetworkManager configuration for the Kali guest's static IPv4 settings.

`K07-static-ip-configuration.png` · 1850 × 745 px

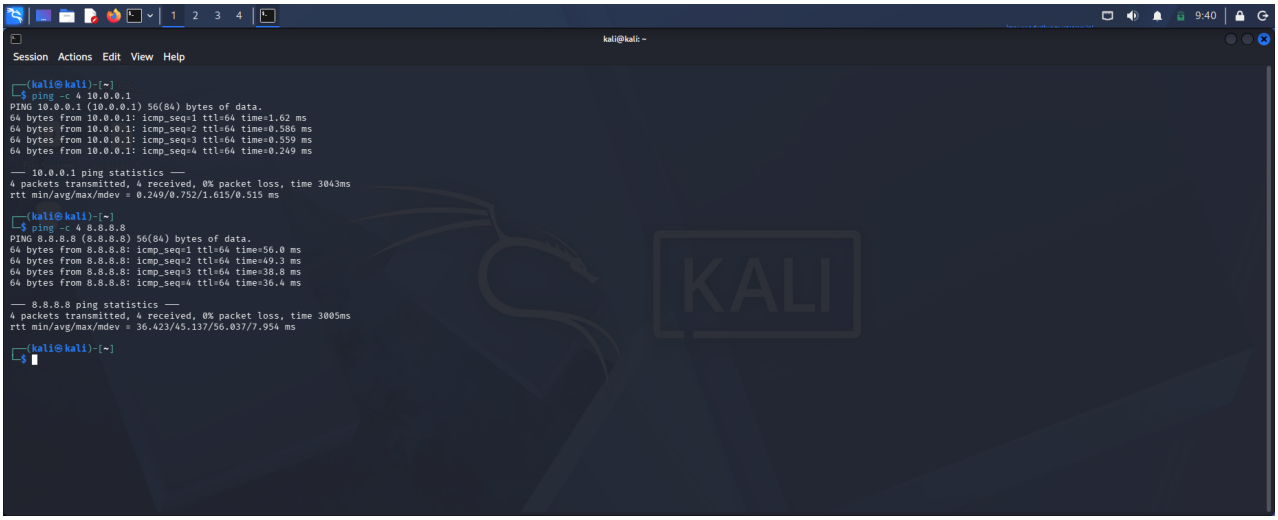
K08 — Interface and routing verification



Recorded interface address and routing table.

`K08-ip-route-device-status.png` · 1850 × 745 px

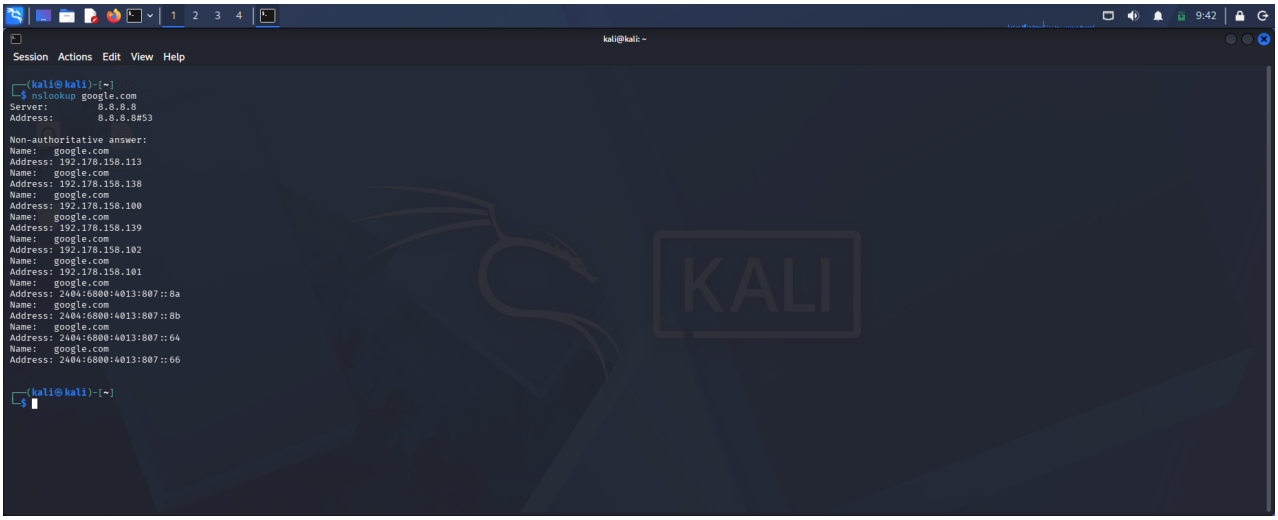
K09–K10 — Connectivity tests



Gateway and external IP connectivity test results.

`K09-K10-connectivity-tests.png` · 1850 × 745 px

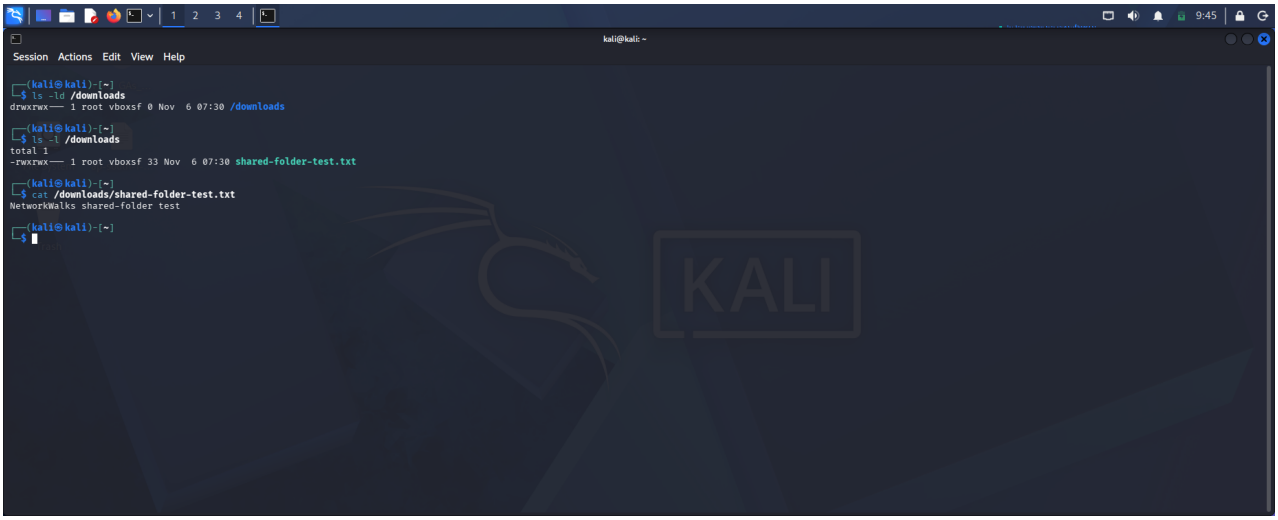
K11 — DNS resolution



Successful DNS lookup using the configured DNS resolver.

K11-dns-resolution.png · 1850 × 745 px

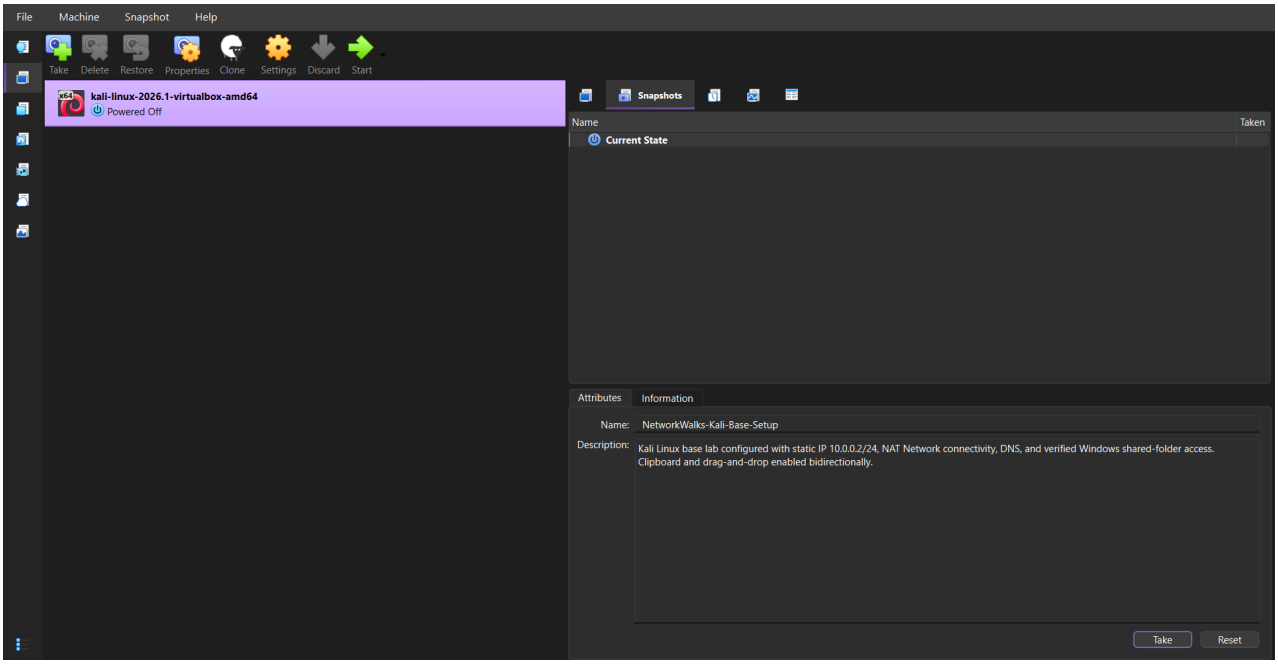
K12 — Shared folder access test



Successful read of the shared-folder test file from Kali.

`K12-shared-folder-access.png` · 1850 × 745 px

K13 — Baseline snapshot



VirtualBox snapshot created after the base Kali lab configuration.

`K13-kali-base-snapshot.png` · 1920 × 991 px

Appendix A — Source and Evidence Inventory

This report is generated from the following repository materials. The Markdown files remain the authoritative editable documentation; this PDF is a formatted report derived from them and the recorded screenshots.

- docs/01-objectives.md
- docs/02-prerequisites.md
- docs/03-virtualbox-setup.md
- docs/04-kali-configuration.md
- docs/05-network-configuration.md
- docs/06-verification.md
- docs/07-troubleshooting.md

Evidence directory

evidence/screenshots/kali/ — screenshot files K01 through K13 (14 image files; K03 is split into three images and K09/K10 share one image).

Appendix B — Scope and Safe Use

This environment is intended for authorized cybersecurity education, controlled lab exercises, and defensive learning. Testing against systems or networks without explicit permission is outside the scope of this project. Keep intentionally vulnerable targets isolated and use only lab assets for practical exercises.